

EXTENSION OF THE MULTI-EGO STRUCTURE-BASED FORCE FIELD TO LIPID MEMBRANES

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ABSTRACT

Membranes are fundamental structures of living systems and, despite an apparent simple composition, still represent a major challenge for state-of-the-art modeling. Membrane internal dynamics and the large size of realistic membranes hamper the possibility of a “computational microscope” in the context of molecular dynamics simulations at atomic resolution. One of the most promising innovations to fill the size and time gaps in modeling is represented by the development of accurate coarse-grained (CG) models, which have recently enabled the simulations of larger membrane systems over longer timescales^{1,2,3}.

In the context of simplified models, we developed multi-eGO, a structure-based/data-driven model which combines atomistic resolution with enhanced computational efficiency by representing only the heavy atoms and implicitly treating the solvent. Following a Bayesian approach, we update a transferable prior with system-specific information learned from training MD simulations. Bonded interactions are transferable while non-bonded interactions are derived from structural data and described using the Lennard-Jones potential. This framework has been shown to successfully represent complex molecular processes at a fraction of the computational cost of all-atom simulations^{4,5}. Here, we extend multi-eGO to lipid systems to describe membrane structure and dynamics.

Representative lipid bilayers were simulated using all-atom MD to generate reference and training data. The framework was extended through introducing lipid-specific parameters and refining bonded and non-bonded interactions. The resulting force field was used to perform POPC membrane simulations, which successfully models membrane properties, such as lipid

packing, membrane thickness and acyl chain ordering, in good agreement with all-atom simulations. This work represents the first step towards the extension of multi-eGO framework to a diverse set of molecules to faithfully represent phenomena of biological interest at larger size and timescale .

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